

# Project Atlas — Delivery Team Pack

Part 1: Team Brief · Part 2: Requirements Digest · Demo Day 15 September

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Issued        | 28 August 2026 · Owner: Sawan                                                                                                             |
| Applies to    | Assigned senior developer(s), senior QA, and office operations                                                                            |
| Hard deadline | Tuesday <b>15 September, 08:00</b> — external stakeholders at our office. Everything below is finished, rehearsed and frozen before then. |

**Confidentiality:** This engagement is in final evaluation. The client's identity is need-to-know and is not named anywhere in this pack. Part 2 of this document contains the complete technical requirements — no other documents are needed to begin. Do not discuss the project outside the assigned team, and nothing on social media — no posts, no "exciting things coming" hints.

## 1 · Context — what this is and why it matters

We are one of a small number of firms being evaluated for a large greenfield build: a **B2C multi-vendor marketplace for the Brazilian automotive aftermarket**, where each franchise store of a national network becomes a seller on a shared storefront. The client has issued a formal Functional Requirements Document (**FRD v1.1**) with stable requirement IDs; our written proposal is due **4 September**; and on **15 September** the client's technical leadership visits our delivery center for a three-hour working session — "Demo Day."

The project is Laravel 12 / PHP 8.4 with a React 19 + TypeScript front end, heavy on queues, multi-tenancy and integration with the client's existing middleware (referred to throughout as "the Hub"). It is exactly the kind of work this team is built for. What we build and how we present in the next two weeks decides whether we win it.

## 2 · Workstream A — Allocation Engine Simulator

**Owner:** Senior Developer 1 · **Effort:** time-boxed 3–4 days · **Feature-complete:** 10 September

The FRD's most bespoke component is §7.2: when several stores can serve the same buyer, demand is *not* won by the cheapest or closest — it is split between eligible sellers in proportion to a 0–5 operational score, subject to a price guard. No off-the-shelf marketplace does this. We are walking into Demo Day with it **already running**. This is a demo, not production code: seeded data, fastest stack wins (Laravel 12 + Livewire or React, SQLite/MySQL — dealer's choice), and the time-box is real.

### Required behaviours

- Score-weighted allocation (FR-GEO-003).** Seed with the exact worked example from the FRD (reproduced in Part 2 §7.2) — three stores, scores 5 / 4 / 3, expected shares 41.7% / 33.3% / 25%. Use those numbers precisely; that detail matters.
- Price guard with a live tolerance slider (FR-GEO-010).** Only sellers within a configurable tolerance of the lowest eligible price enter the draw. The slider must show the business lever working: wide tolerance → score decides; narrow → price decides; zero → "cheapest always wins" plus the required admin warning. This slider is the heart of the demo — make it smooth.
- Simulated purchase stream (FR-GEO-004).** Run N purchases and chart realised share converging on target share, with the tolerance visible. A simple live chart is enough.
- Session stability (FR-GEO-005).** Reloading the product page in the same session keeps the same featured seller.

5. **Decision audit (FR-GEO-006 / 011).** Click any allocation decision and see: who was eligible, their scores, the lowest price in region, the tolerance in force, who the price guard excluded, and why the winner won. "Auditable, not a black box" is a direct quote from the FRD — build to that sentence.

**Deployment:** runs fully offline on the demo machine (no wifi dependency on the day), *and* deployed to a private URL the visitors can take away. Expect them to grab the keyboard — it must survive being driven by someone else.

### 3 · Workstream B — QA Live Demo

**Owner:** Senior QA · **Effort:** ~1 day prep · **Ready:** 10 September

- A **ten-minute live run** of a real Playwright regression suite: many recorded flows executing, one deliberately failing, the failure flagging itself. Show the suite the way we actually work — Pest for unit/feature, Playwright for E2E, CI as the quality gate.
- **Use an internal or demo application only.** Never show a client's system without written permission — if in any doubt, it does not go on screen.
- Prepare and rehearse a crisp answer to: *"You join a team with 20 pull requests stuck waiting for one QA. What do you do in week one?"* Triage order, what gets automated first, tooling, how the backlog reaches zero. The client team lives this problem daily; your answer should sound like a plan, not a philosophy.

### 4 · Workstream C — Requirements mastery

**Owners:** both assigned designates · **Done:** 5 September

- Read **Part 2 of this pack end to end.** It is the complete requirements digest for the project, organised by the client's own FRD section numbers and requirement IDs (FR-XXX-NNN). Those IDs are the shared vocabulary on the 15th — when a visitor says "\$7.2" or "FR-GEO-010", you should know exactly what is meant.
- Each designate brings **two or three written technical observations or questions** to the 11 September dry run. Genuine ones — things you would actually raise in sprint one.
- The developer must know **§15 (the Hub integration contract)** cold: idempotent inbound operations, per-seller scoping, webhooks out, stable error codes, the readiness checklist.
- Both designates must be fluent on the open question we have already flagged to the client: **seller-assembled kits (FR-CAT-045) vs FR-LST-001 and FR-PRC-001** — if all listings and prices flow from the Hub, where is a kit defined and priced? Expect this to come up on the 15th; the person who looks blank at a point our own company raised undoes the whole morning.
- The expected screen inventory is in Part 2 §18 — that is the UI surface area. A visual export of the designs may follow separately if needed.

### 5 · Workstream D — Estimate review

**Owners:** Senior Developer 1 with Sawan · **Done:** 3 September

Before the proposal goes out on the 4th, the developer reviews the module-by-module effort table against FRD v1.1 and **owns it technically.** On the 15th the visitors will look at the engineer — not at Sawan — and ask whether the timeline is realistic. The answer must be a confident yes with a reason. If any module feels wrong, say so now; changing an estimate this week costs nothing, changing it in sprint two costs the relationship.

### 6 · Demo Day — Tuesday 15 September

- **08:00–11:00 at the delivery center.** External stakeholders attending. Full team at desks by **07:45** — a delivery center with empty chairs sells nothing.

- **Speaking roles:** Senior Developer 1 drives the simulator; Senior QA drives the test demo; Sawan drives everything else. Everyone else works normally and is ready to say hello.
- **Language discipline:** our visitors are not native English speakers. Slow pace, short sentences, no idioms, pause after key points. This is rehearsed on the 11th, not improvised on the 15th.
- **Be ready to talk about yourselves:** the designates should be comfortable walking through their background, stack and recent work in a few unhurried minutes.
- **Room (ops, tested by 11 September):** demo machine *plus* a backup laptop, both with the simulator loaded and working offline; display/projector and cables tested with the actual machines; a light welcome screen; chai and breakfast ready for 08:00. Working session, not a ceremony — keep it simple.

## 7 · Materials to prepare (by 11 September)

- **One-page profiles** for Senior Developer 1 and Senior QA, plus one alternate each: stack, years, notable work, certifications. Clean, printable, no client names from past work without permission.
- **Simulator one-pager:** what it shows, which FRD requirements it maps to, and the private URL.

## 8 · Deadline summary

| Date       | Milestone                                                           | Owner                    |
|------------|---------------------------------------------------------------------|--------------------------|
| Wed 3 Sep  | Effort table reviewed and owned                                     | Senior Dev 1 + Sawan     |
| Sat 5 Sep  | Part 2 (Requirements Digest) read end to end; written notes drafted | Both designates          |
| Thu 10 Sep | Simulator feature-complete · QA demo ready                          | Senior Dev 1 · Senior QA |
| Fri 11 Sep | Team-only dry run (recorded) · room/AV tested · materials done      | All + Ops                |
| Mon 14 Sep | Final dry run with Sawan · full freeze — fixes only, nothing new    | All                      |
| Tue 15 Sep | <b>Demo Day, 08:00</b> — everyone in, everything running            | All                      |

Questions, blockers, or anything that threatens a date on this table: raise it to Sawan the day you see it, not the day it's due. Two weeks is enough time to do this well and no time at all to do it twice.

## Part 2 — Requirements Digest

Complete technical substance of the client's FRD v1.1 · identifying details removed · requirement IDs preserved

This digest reproduces every requirement of the client's FRD v1.1 in condensed form. Section numbers and requirement IDs match the client's document exactly and are the authoritative vocabulary for this project. Priorities: **MUST** (MVP) · **SHOULD** (may slip to Phase 2 if it materially speeds launch — priced separately) · **Phase 2** (deliberately deferred) · **OUT OF SCOPE** (decided against; no workaround to be proposed). All seven previously-open business decisions are settled; two narrow items remain open (§16.8).

### §1–2 · Project context

The client is a national Brazilian franchise network of automotive parts stores. We are building a **B2C multi-vendor marketplace**: a consumer storefront where each franchise store is a seller (tenant) and the buyer is an end consumer identified by CPF. **Mercado Livre is the reference model** — where a requirement is ambiguous, current Mercado Livre behaviour for a Brazilian consumer is the tiebreaker, except for the two structural differences below (§3.3). The document specifies behaviour only; stack, infrastructure and non-functional targets are ours to propose.

**The two structural differences from Mercado Livre (§3.3)** — where most of the non-obvious effort sits: (1) all sellers are franchisees of the same brand, so open price war on the product page is a business risk, not a feature — hence the geographic allocation and score-based demand split of §7, which has no Mercado Livre equivalent; (2) vehicle fitment is the primary navigation axis — buyers search "brake disc for my 2020 Corolla", so fitment is a first-class search dimension, not a filter afterthought.

### §3 · System context and non-negotiable boundaries

Three systems: **seller ERPs** (many, modern and legacy) → **the Hub** (client-owned middleware, already built; single operational source of truth for everything belonging to a seller) → **the B2C marketplace** (our scope). Orders flow back from marketplace to Hub. A sibling B2B portal already exists; its credit/invoice model must not leak into B2C.

- **FR-CTX-001 (MUST)**. Never integrate directly with any seller ERP. All seller data enters and leaves exclusively through the Hub.
- **FR-CTX-002 (MUST)**. No authoritative product/stock/price registry that ignores the Hub — the marketplace reflects what the Hub publishes.
- **FR-CTX-003 (MUST)**. No B2B credit limits, invoice-on-account, ERP bank slips, or price-per-CNPJ. The channel is gateway\_only: the payment gateway charges the buyer; the Hub never authorises payment.
- **FR-CTX-004 (MUST)**. The channel is registered in the Hub under a b2c channel code (value withheld here). Every exchanged event belongs to exactly one seller.

### §4 · Actors and roles

Actors: consumer (CPF, registered or guest) · seller operator and seller admin (franchisee staff/owner) · client-side catalog moderator, operations/dispute agent, and commercial admin (back-office) · the Hub and seller-side integrators as system actors.

- **FR-ROL-001 (MUST)**. Role-based access control, assignable per user, scoped per seller and per branch.
- **FR-ROL-002 (MUST)**. Strict tenant isolation between sellers. Cross-tenant object access (IDOR) is treated as a critical defect.

## §5 · Catalog and product data

**Ownership model:** a client-owned base catalog (multi-million-SKU scale) is the foundation. Sellers never create free-form products; they contribute to and list against the base catalog: sellers → integrator → moderation → base catalog → B2C search.

- **FR-CAT-001 (MUST).** Maintain the base catalog of products and SKUs, importable and updatable in bulk from the client's catalog source.
- **FR-CAT-002 (MUST).** Match process on every incoming seller product: attach to an existing base product rather than duplicate.
- **FR-CAT-003 (MUST).** Match uses at minimum EAN/GTIN, manufacturer part number, brand + part number, normalised description. Outcomes: automatic match · candidate match (moderation) · no match (new product candidate).
- **FR-CAT-004 (MUST).** Non-matches enter a moderation queue: approve as new, merge, or reject with reason code.
- **FR-CAT-005 (MUST).** Seller contributions can enrich the base product (image, dimension, description, fitment, attribute), subject to moderation, with a full audit trail of who contributed what and when.
- **FR-CAT-006 (MUST).** Per-seller contribution dashboard: accepted, pending, rejected — with the moderator's reason on every rejection. (Upgraded from SHOULD; it is the substrate of the recognition scheme.)
- **FR-CAT-060 (SHOULD).** Non-financial recognition layer: seller ranking by accepted contributions over a configurable period, a store-page badge, and an export for recognition outside the platform. Two hard constraints: enrichment must never feed the seller score (§8) or demand allocation (§7.2), and must never vary commission.
- **FR-CAT-061 (MUST).** Moderation queue exposes throughput metrics — pending count, oldest-item age, average time to decision, decisions per moderator — so moderation capacity can be sized.
- **FR-CAT-007/008 (MUST).** Product record: name, short/long description, brand, category, status, ordered images with a primary, typed category-dependent attributes, video, associated SKUs. SKU record: internal code, EAN/GTIN, weight, dimensions, variant axes, packaging, image links.
- **FR-CAT-009 (MUST).** Persist Hub identifiers on every entity (seller, product, SKU, branch, order) and return our `external_id` for every listing and order.
- **FR-CAT-010–012 (MUST).** Marketplace owns its own category tree and brand registry; both exposed via queryable API (search/autocomplete) for the Hub's mapping; listings submitted with marketplace codes, never ERP free text.
- **FR-CAT-013 (MUST).** Per-category mandatory/optional attributes with types and permitted values, exposed via API so the Hub validates before publishing.
- **FR-CAT-014 (MUST).** Unmapped fallbacks must not fail silently: reject with stable machine codes (`CATEGORY_NOT_FOUND`, `BRAND_NOT_FOUND`, `MISSING_REQUIRED_FIELD`, `INVALID_ATTRIBUTE_VALUE`) plus a human-readable message.
- **FR-CAT-020–022 (MUST).** Vehicle fitment as a many-to-many between SKU and vehicle definitions (make, model, year range, engine, trim); bulk-importable, back-office maintainable, usable as search dimension, filter and PDP element; one SKU may fit multiple manufacturers.
- **FR-CAT-023 (SHOULD).** Licence-plate lookup via a pluggable external vehicle-data provider: resolve plate → vehicle → pre-filtered catalog.
- **FR-CAT-024 (MUST).** Vehicle segments as top-level navigation: cars & pickups, trucks & buses, motorcycles; plus light line, heavy line, agricultural.
- **FR-CAT-030–032.** Synonym dictionary at catalog level feeding search without polluting display names (MUST); regional scoping of synonyms (SHOULD); analytics surfacing high-volume zero/poor-result queries (SHOULD). Brazilian parts vocabulary is strongly regional.

## §5.5 · Kits (settled: seller-assembled only at launch, bundle-product semantics)

Mental model: a kit is *one commercial object to the buyer* and *a set of independent line items to the order, the fiscal invoice (NF-e) and the ERP* — the bundle product in the Magento / Adobe Commerce sense.

- **FR-CAT-040 (MUST).** Kit = saleable product of component SKUs in defined quantities, with its own PDP, price, images and listing lifecycle.
- **FR-CAT-041 (MUST).** All-or-nothing availability at the fulfilling branch: never a partial kit, a substitution, or an incomplete shipment.
- **FR-CAT-042 (MUST).** In the cart, one line: components visible but bound — not removable, substitutable or individually re-quantified; quantity applies to the kit.
- **FR-CAT-043 (MUST).** In the order, components become independent line items with own SKU, quantity, and unit price allocated from the kit price; the Hub payload itemises them; the ERP picks and invoices them; the NF-e lists them — while the order retains the kit's commercial identity.
- **FR-CAT-044 (MUST).** Order page shows kit + component breakdown together, so the buyer recognises the itemised invoice. Stated requirement, not a UI nicety.
- **FR-CAT-045 (MUST).** At launch, kits are defined by the individual seller from their own SKUs, fulfilled from one branch, and pass mandatory moderation. (*Our flagged question to the client: where exactly is a kit defined and priced, given FR-LST-001 and FR-PRC-001 route listings and prices through the Hub.*)
- **FR-CAT-046 (Phase 2).** Client-assembled standard kits published to all eligible sellers; buyer-facing dynamic kit builder with bundle pricing.
- **FR-CAT-047 (MUST).** Incomplete kit → listing unavailable; allocation applies among sellers whose equivalent kits are complete; if none can fulfil for that postal code, the kit page shows unavailable and offers components individually.
- **FR-CAT-048 (MUST).** Components are the stock-bearing entities; kit availability is computed from components at read time; the kit holds no independent balance.

## §6 · Listings, offers and the PDP

The client calls this "the single most important design decision in the project."

- **FR-PDP-001/002 (MUST).** One product page per product — never one page per seller listing. Each seller's offer (price, stock, origin branch, shipping, delivery estimate) attaches to the shared product record.
- **FR-PDP-003 (MUST, settled display mode).** The PDP shows a featured offer ("buy box") chosen by §7.2 allocation, plus a collapsed "other stores near you" section. This mode is fixed — no configurability to build.
- **FR-PDP-009 (MUST).** The collapsed list is *never* ordered by price and offers no price sort. Ordered by proximity, seller score as tiebreaker; price displayed per entry but never the comparison axis. Non-negotiable at implementation time — a price-sorted collapsed list would bypass the allocation engine.
- **FR-PDP-010 (MUST).** Collapsed section closed by default with a count; each entry shows store, distance, score, delivery estimate; buyer may deliberately choose a non-featured store, the platform records it, and FR-GEO-004 accounts for it in realised-vs-target share.
- **FR-PDP-004 (MUST).** PDP shows: name, brand, fitment, image gallery, video when present, technical attributes, description, price + instalments, PIX price, shipping quote by postal code, delivery estimate, stock signal, seller identity, seller score, buyer questions, reviews.
- **FR-PDP-005/006 (MUST).** Title = product name + brand + vehicle application, where the application shown is the buyer's selected/searched one; one product presents different application labels in different contexts without duplicate products.
- **FR-PDP-007 (SHOULD).** One video per product minimum; "has video" usable as a ranking signal.

- **FR-PDP-008 (MUST).** "Sold and shipped by" attribution on the featured offer and every collapsed entry; never present a franchisee-fulfilled offer as the client's own. (Under settled Model B, the franchisee is merchant of record — this is a factual statement.)
- **FR-LST-001–004 (MUST).** Listings exist only when the Hub publishes them for that seller; create/update/pause/unpublish/remove driven by the Hub, returning `external_id`; states observable by the Hub: active, inactive/paused, under\_review, rejected/suppressed; Hub-side pause/block reflected on the storefront within the agreed propagation window.
- **FR-LST-005 (MUST).** Our own listing quality gate: title/description length limits, minimum image count (1, 3+ recommended), image formats/dimensions, mandatory weight/dimensions, mandatory EAN where the category demands, category-mandatory attributes — rejections with stable codes.
- **FR-LST-006/007 (MUST).** Moderation mandatory for all sellers, including trusted ones (stricter than B2B by design); marketplace-side rejection/suppression notifies the Hub so it blocks re-publication.

## §7 · Search, ranking and geographic demand allocation

- **FR-SRCH-001 (MUST).** Full-text search across name, brand, part number, EAN, synonyms, fitment — typo tolerance, Brazilian-Portuguese stemming, autocomplete.
- **FR-SRCH-002/003.** Fitment-first search: make → model → year → version, persisted across the session ("my garage") (MUST); multiple saved vehicles per account, switchable from the header (SHOULD).
- **FR-SRCH-004/005 (MUST).** Facets: category, brand, price range, shipping (free / arrives today / within 24h), delivery time, vehicle line, own brand, rating. Sorting: relevance, price asc/desc, best rated, newest; grid and list views; page-size selection.
- **FR-SRCH-006/007 (MUST).** One card per product (not per offer), showing the best available offer per allocation; only stock-available, location-serviceable listings shown as purchasable — out-of-region/out-of-stock handled explicitly per configuration.
- **FR-SRCH-008/009 (SHOULD).** Client-configurable weighted ranking (text relevance, fitment exactness, seller score, price competitiveness, stock depth, delivery time, has-video, conversion history, campaign boost); search analytics dashboard.

### §7.2 · Geographic allocation and score-weighted split — the bespoke engine

Principle: when several stores can serve the same buyer, demand is *not* won by cheapest or closest — it is split among eligible sellers in proportion to seller score, subject to the price guard. No Mercado Livre equivalent exists.

- **FR-GEO-001/002 (MUST).** Each branch has a service area (postal-code ranges and/or radius), client-configurable and overridable; given a buyer location, compute eligible sellers = service area covers buyer  $\wedge$  product in stock at a serving branch.
- **FR-GEO-003 (MUST).** Featured offer assigned by score-weighted allocation: each eligible seller's share = score ÷ sum of eligible scores. Worked example: Store A (5), Store B (4), Store C (3) → 41.7% / 33.3% / 25.0%.
- **FR-GEO-004 (MUST).** Allocation is stateful and balanced over time — not a per-request random draw. Realised share tracks target within a configurable tolerance; realised-vs-target exposed to the client.
- **FR-GEO-005 (MUST).** Session-stable: same featured seller on reload/return within a session.
- **FR-GEO-006 (MUST).** Auditable, not a black box: per product and postal code — who was eligible, their scores, why the winner won. Franchisees will challenge decisions.
- **FR-GEO-007/008 (SHOULD).** Configurable no-coverage fallback (nothing / nearest with extended shipping / client-fulfilled offer); configurable eligibility floor (exclude or cap low-score sellers).
- **FR-GEO-009 (MUST).** Other sellers' offers visible via the collapsed section (see FR-PDP-003/009/010) — ordering rule is part of the settlement.
- **FR-GEO-010 (MUST).** **Price guard:** compute the lowest eligible price; only sellers within a configurable tolerance of it enter the score-weighted draw. Others keep their listing and collapsed-list presence but cannot hold the buy

box. Tolerance set globally, overridable per category. Wide → score decides; narrow → price decides; zero → "cheapest always wins" (permitted, but the platform must warn the administrator).

- **FR-GEO-011 (MUST).** Audit records the guard: eligible sellers, lowest price, tolerance in force, who was excluded, and the draw among the rest. "Your price was outside the band" must be demonstrable.
- **FR-GEO-012 (MUST).** Realised-vs-target reporting separates share lost to the price guard from share lost to a low score — a pricing decision vs a 30-day operational record; the client must be able to tell franchisees apart.

### §7.3 · Regions (settled: one primitive, campaigns at launch)

- **FR-RGN-001 (MUST).** A region = named set of postal-code ranges (postal code is the only geographic key the Hub, shipping quote and buyer address share). A state is a region whose ranges cover it.
- **FR-RGN-002 (MUST).** One region model, not two: FR-GEO-001 service areas bind branches to these same regions; allocation eligibility, price guard, campaigns, synonyms and delivery promises all resolve the buyer's postal code to the same region set.
- **FR-RGN-003 (MUST).** Regions may overlap and are versioned — "which region was this buyer in when this order was placed" must be answerable after the map changes.
- **FR-RGN-004 (MUST).** Regional campaigns at launch: promotional price, free-shipping threshold, homepage banner or category boost — targetable to regions with a validity window, visible only to buyers resolving into them.
- **FR-RGN-005 (MUST).** Regional synonym scoping (FR-CAT-031) resolves against this same entity.
- **FR-RGN-006.** Regional content is Phase 2 on the same primitive. Regional delivery promises are already covered by FR-SHP-004. **Regional assortment is OUT OF SCOPE** (excluded, not deferred — it collides with FR-SRCH-007 and FR-GEO-002).

## §8 · Seller score

Drives allocation and is displayed to buyers as a trust signal. Score = (late-rate grade × 0.40) + (cancellation-rate grade × 0.30) + (mediation-rate grade × 0.30), each rate over a rolling 30-day window, converted to 0–5 grades by the band table below (initial configuration, not constants).

| Grade             | 5      | 4          | 3           | 2            | 1            | 0       | Weight |
|-------------------|--------|------------|-------------|--------------|--------------|---------|--------|
| Late rate         | ≤5.00% | 5.01–8.00% | 8.01–10.00% | 10.01–15.00% | 15.01–20.00% | ≥20.01% | 40%    |
| Cancellation rate | ≤2.00% | 2.01–3.00% | 3.01–4.00%  | 4.01–5.00%   | 5.01–10.00%  | ≥10.01% | 30%    |
| Mediation rate    | ≤1.00% | 1.01–2.00% | 2.01–3.00%  | 3.01–4.00%   | 4.01–5.00%   | ≥5.01%  | 30%    |

Late rate = (delivered late + in-progress-and-late) ÷ (delivered + in-progress-and-late) × 100. Cancellation and mediation rates = count ÷ delivered × 100. Worked example: 5.00% / 2.56% / 1.03% → grades 5 / 4 / 4 → score 4.4.

- **FR-SCR-001–004 (MUST).** Compute per seller (and per branch where configured), recalculated at least daily; weights, bands, window and frequency configurable without deployment; insufficient volume → "not qualified" (threshold and treatment configurable); full score history auditable for any past date.
- **FR-SCR-005 (MUST).** Seller panel shows current score, each component rate, band, distance to next band, and the individual orders driving each metric.
- **FR-SCR-006 (SHOULD).** Buyer-facing reputation display (0–5 thermometer style with supporting signals).
- **FR-SCR-007 (MUST).** Correct cancellation attribution: buyer- or payment-caused cancellations never count against the seller; rules configurable and auditable.
- **FR-SCR-008 (SHOULD).** Alerts on band deterioration (to seller) and floor breach (to client operations).

- **FR-SCR-009 (MUST).** The score is a reusable, configurable decision input — a named signal consumable, independently switchable and weightable, by: allocation, search ranking, campaign eligibility, client-funded shipping-subsidy eligibility, premium placement.
- **FR-SCR-010 (MUST).** Settled policy: de-prioritise where the consequence is visibility (ranking, allocation share); exclude where the consequence is the client spending money on the seller (campaigns, funded shipping subsidy). Exclusion threshold separate from the allocation floor.
- **FR-SCR-011 (MUST).** Any exclusion is visible to the affected seller with the metric, threshold, and the route back — never a silent absence.
- **FR-SCR-012 (MUST).** No input other than the three operational rates may enter the score — not enrichment, volume, tenure, or campaign participation.

## §9 · Pricing

- **FR-PRC-001/002 (MUST).** Price comes from the Hub (final value to publish; we never recalculate from base cost). Single channel price table — no price-per-customer, no price-per-CNPJ.
- **FR-PRC-003 (MUST).** Promotions arrive from the Hub as a special price with a validity window; we apply and expire them; no parallel promotion engine.
- **FR-PRC-004 (MUST).** Display: from/for pricing, discount badges, PIX price, instalment terms.
- **FR-PRC-005 (OUT OF SCOPE).** Price per vehicle application — not built in any phase; no workaround (explicitly including the two-SKUs-for-one-part trick, which would double-count one physical shelf under FR-INV-001).
- **FR-PRC-006 (SETTLED).** Regional pricing without a regional price table: it emerges from which seller wins the featured offer in a region, bounded by the FR-GEO-010 price guard. Hub price contract unchanged. (Where "regional pricing" is really a delivery-cost difference, FR-SHP-001 already answers it.)
- **FR-PRC-007 (MUST).** Price changes never retroactively alter placed orders; every order line stores a price snapshot.
- **FR-PRC-008 (SHOULD).** Client-level campaign pricing distinguishable from seller price; interaction with Hub price explicit and auditable.
- **FR-PRC-009 (Phase 2).** Client-subsidised promotional price for exceptional events (e.g. Black Friday) as the only permitted deviation from Hub price: seller made whole at Hub price; subsidy recorded per order line (Hub price, promo price, subsidy, campaign id) and included in Hub payload and settlement export; Hub remains price of record; time-boxed, region-targetable (FR-RGN-004), auto-expiring; auditable end to end.

## §10 · Inventory

- **FR-INV-001/002 (MUST).** Stock per SKU + branch; available = on-hand – reserved. Stock arrives from the Hub; never invent, estimate or extrapolate a balance.
- **FR-INV-003 (MUST).** Zero stock: Hub may instruct auto-pause; replenishment never auto-republishes — the seller decides in the Hub. Respect the asymmetry.
- **FR-INV-004 (MUST).** Idempotent, out-of-order-safe stock updates: a stale update never overwrites a newer one.
- **FR-INV-005 (MUST).** Checkout reserves stock for the payment attempt and releases on failure/expiry — two buyers cannot buy the last unit.
- **FR-INV-006 (SHOULD).** Configurable "last units" display and a hide-below safety threshold.

## §11 · Cart, checkout and payment

- **FR-CRT-001–004 (MUST).** Multi-seller cart grouped by seller with per-seller shipping/estimates; splits into one order per seller at checkout (independently fulfilled, tracked, invoiced, cancellable) while the buyer sees one purchase; persistence across devices for logged-in buyers, browser-session for guests; revalidate price/stock/eligibility at checkout entry with warnings before payment.

- **FR-CHK-001–005.** CPF + email registration, password recovery, guest checkout capturing CPF/name/contact (MUST); social login (SHOULD); address book with postal-code autofill, multiple addresses, separate delivery/billing (MUST); per-seller-group checkout display with totals incl. any client-funded discount (MUST); Brazilian consumer-law compliance: clear total before payment, confirmation with commercial terms, withdrawal/return information (MUST).
- **FR-PAY-001/002 (MUST).** gateway\_only; methods: credit card with instalments, PIX, boleto (ours, never the ERP's).
- **FR-PAY-003–005 (MUST).** Status approved/pending/rejected sent to the Hub with transaction id, method, instalments, amount (audit only); never store raw card data (gateway tokenisation); refunds, partial refunds and chargebacks reflected in the order and the Hub payload.
- **FR-PAY-006 (MUST).** Client commission calculated at checkout and recorded on the order (rate, base, amount), included in the Hub payload. Settled Model B: the franchisee sells at their own price and pays the client a commission.
- **FR-PAY-007 (MUST).** Gateway split at the moment of payment for every method that supports it (in practice card and PIX): commission retained, seller remainder released to the seller's own account. Launch requirement.
- **FR-PAY-008 (MUST).** Non-split methods (boleto expected) settle by reconciliation: full amount received centrally; finance settles against the export. Which methods fall on which path is confirmed against the chosen gateway.
- **FR-PAY-009 (MUST).** Settlement reconciliation export per seller per period: orders, gross, commission, gateway fees, refunds/chargebacks, amounts released by split, amounts pending manual settlement, net payable — visible in the seller panel and machine-readable for finance; same figures on both sides.
- **FR-PAY-010 (MUST).** Per-order settlement state: split\_at\_gateway · pending\_manual\_settlement · settled · reversed — visible to seller and client finance.
- **FR-PAY-011 (MUST).** Seller transaction view (Mercado-Pago-style): transactions, status, fees, expected release date, settlement history.

## §12 · Shipping

- **FR-SHP-001/002 (MUST).** Marketplace-quoted shipping at postal-code level on PDP, cart and checkout, per seller group — cost, carrier/modality, delivery estimate — considering origin branch, destination, weight, dimensions, declared value.
- **FR-SHP-003 (MUST).** At minimum: Correios, a configurable carrier table (client- or seller-maintained), and free-shipping rules by value threshold, region, campaign.
- **FR-SHP-004/005 (SHOULD).** "Arrives today" / "within 24h" local-delivery promises (rides on §7.2 proximity — a differentiator for a physical-store network); click & collect with store selector and collection workflow.
- **FR-SHP-006–008 (MUST).** Shipping amount/modality/lead time/address in the Hub order; label generation and printing from the seller panel plus reverse-logistics voucher for returns; tracking capture (panel or API) with buyer-facing status updates.

## §13 · Orders, fulfilment and the seller panel

Once the order reaches the Hub, the Hub and the ERP own fulfilment; the marketplace owns the buyer's experience of it.

- **FR-ORD-001 (MUST).** Normalised order to the Hub: seller + origin branch, line items keyed by Hub SKU id with quantity and price snapshot, delivery/billing addresses, shipping (amount, modality, lead time), payment (status, method, instalments, transaction id, amount), commission record, buyer identity (CPF, name, contact).
- **FR-ORD-002/003 (MUST).** Idempotency key on submission — one checkout never produces two Hub orders under any retry; persist the returned hub\_order\_id as the correlation key.
- **FR-ORD-004 (MUST).** Consume Hub status updates — pre-processing, sent-to-ERP, invoiced (NF issued), shipped, delivered, cancelled, returned — updating buyer view and notifications.

- **FR-ORD-005 (MUST).** Marketplace-initiated cancellation is valid only if the Hub still permits the transition: ask, don't assume; present the buyer an honest outcome on refusal.
- **FR-ORD-006/007 (MUST).** NF-e attached by integration or panel upload, downloadable by the buyer; full order timeline visible to buyer, seller and client operations — every transition timestamped and attributed.
- **FR-SEL-001–004 (MUST).** "My sales" list with filters, search, bulk actions, export; order detail with fulfilment checklist (issue NF → print label → dispatch → add tracking); NF upload; label + reverse-voucher printing.
- **FR-SEL-005 (MUST).** "My disputes": mediation/claim queue with SLA countdown, message thread, evidence, resolution history.
- **FR-SEL-006 (MUST).** Catalog/stock/price views read-only where the Hub is source of truth, with clear pointers to where changes are made — never offer edits the Hub would overwrite.
- **FR-SEL-007/008 (MUST).** Performance dashboard (score + components, sales, conversion, cancellations, late shipments, open questions); buyer-questions inbox with response-SLA tracking.
- **FR-SEL-009 (SHOULD).** Seller API mirroring panel read/write operations for franchisees with their own IT or integrators (catalog/stock/price still flow through the Hub).
- **FR-SEL-010 (MUST).** Multi-branch: one login, per-branch filtering and permissions.

## §14 · Post-sale: questions, reviews, disputes

- **FR-MSG-001–003 (MUST).** PDP buyer questions answered by the seller, published once answered, client-moderated; Hub notified of events the seller needs outside the marketplace (e.g. new question); order-scoped post-purchase buyer ↔ seller messaging.
- **FR-MSG-004/005 (MUST).** Verified-purchase-only product reviews with moderation; delivered-order-derived seller rating, distinct from the §8 operational score but displayable alongside it.
- **FR-DSP-001–004 (MUST).** Buyer claims (not received / different / defective / wrong) with evidence; escalation to client mediation on SLA breach (feeds the mediation rate); client arbitration workspace (full context, history, evidence, decision, outcomes: refund, partial refund, return, reject); returns tied to the reverse-logistics voucher with status visible to all three parties.
- **FR-DSP-005 (SHOULD).** Configurable SLAs and automatic escalation timers per claim type.

## §15 · Hub integration — the contract

The Hub implements a Channel Adapter for this channel (same pattern as the existing B2B adapter); we expose a stable API. Payloads/OpenAPI/auth are agreed jointly during design. Envelope reference: event id, seller\_id, event type, complete state (not deltas), fast acknowledgement (202; 200 for duplicates).

- **Hub → Marketplace:** catalog, stock, price, publication, order status, seller provisioning — authenticated REST/webhooks, asynchronous, idempotent processing.
- **Marketplace → Hub:** new orders, payment updates, listing status changes (moderated/suppressed), buyer questions — authenticated REST + signed webhooks.
- **FR-INT-001–010 (all MUST).** Platform health endpoint · per-seller connection health · seller provisioning (create/deactivate/reactivate on Hub instruction) · listing create/update/pause/unpublish returning external\_id · inventory sync (with zero-stock pause) · price sync incl. special price with validity window · order handling (Hub acknowledges the normalised order, then drives the workflow) · signed, idempotent webhooks for order created/updated, listing changed, question received · queryable taxonomy (categories, brands, category-mandatory attributes) · stable machine-readable error codes for invalid listings.
- **FR-INT-011–013 (MUST).** Auth: platform-level token and per-seller credentials; shared-database or "native channel without an API" approaches are unacceptable. Every operation scoped to one seller, tenant resolved from the Hub-provisioned link. All inbound operations idempotent and retry-safe — the Hub will retry.

- **FR-INT-014 (MUST)**. Integration observability: per-seller event log, failure reasons, retry state, and replay capability for the client's technical team.
- **Readiness checklist (§15.3) = integration acceptance criteria:** provisioning API · both health endpoints · idempotent async catalog/stock/price ingestion · full listing lifecycle with external\_id · taxonomy exposed · stable rejection codes · orders sent with payment, Hub-SKU-keyed items, address and idempotency key · buyer-facing order/NF/shipping status driven by Hub returns · no ERP access, Hub never treated as optional.

## §16 · Decision record — settled points and what remains open

| Point                                       | Settled outcome                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 16.1 Price per application / regional price | Application pricing OUT OF SCOPE; regional price emerges via the FR-GEO-010 price guard — no regional price table                   |
| 16.2 Buy-box display                        | Featured offer + collapsed "other stores near you", never price-ordered (FR-PDP-009/010) — single mode, no configurability          |
| 16.3 Score reuse                            | Reusable configurable decision input (FR-SCR-009); de-prioritise on visibility, exclude from client-funded benefit (FR-SCR-010/011) |
| 16.4 Remuneration                           | Model B — seller sells at own price, pays commission; seller is merchant of record; Model A will not be built                       |
| 16.5 Regionalisation                        | One region primitive (§7.3); regional campaigns at launch; regional content Phase 2; regional assortment excluded                   |
| 16.6 Kits                                   | Seller-assembled only at launch, bundle-product semantics (FR-CAT-040–048); client kits + dynamic builder Phase 2                   |
| 16.7 Enrichment incentives                  | Non-financial recognition only (FR-CAT-060/061); never feeds score, allocation or commission                                        |

**Still open (§16.8), neither blocking:** (1) choice of payment gateway + confirmation of which methods support instant split in Brazil — client decides with our recommendation, needed before the payment workstream starts; (2) the price-guard tolerance value — client configuration, needed before launch, not before build.

## §17 · Phasing summary

- **Phase 1 (MVP):** catalog + match/moderation, taxonomy API, fitment + fitment-first search, single-PDP-many-offers with collapsed section, eligibility + price guard + score-weighted allocation, region primitive + regional campaigns, score engine + reusable decision input, search with facets/sorting, seller kits with bundle semantics, cart/checkout/CPF + guest, gateway payment (card/PIX/boleto) with commission, gateway split + reconciliation export + settlement states, shipping quote/label/tracking, Hub-synchronised order lifecycle, NF upload/download, full seller panel incl. transactions/settlement, client back-office (moderation, taxonomy, sellers, regions, allocation + price-guard audit), full §15 contract, buyer account, contribution tracking + recognition layer.
- **Phase 2:** licence-plate search, saved garage, click & collect, client-assembled kits + dynamic kit builder, subsidised promotional pricing (FR-PRC-009), regional content, advanced ranking config + search analytics, photo/video reviews, advanced dispute automation/SLA engine.
- **Out of scope:** price per application, regional assortment, Model A, consumer CRM/segmentation, advanced RMA, a logistics engine on the Hub side, generic ticketing, any direct ERP connector, and any general financial module beyond FR-PAY-007–011 (no AP, tax engines, seller credit).

## §18 · Expected screen inventory

- **Storefront:** home with campaign areas · vehicle selector (plate / cars & pickups / trucks & buses / motorcycles) · results grid/list with facets · PDP with offers, fitment, gallery, video, questions, reviews · seller store page · cart · checkout (identification, address, shipping, payment, review) · order confirmation · buyer account (profile, addresses, garage, orders + tracking, NF download, returns, claims, questions, reviews).
- **Seller panel:** login + seller switcher · dashboard (score + sales) · my sales (list, detail, fulfilment actions) · NF upload · label + reverse-voucher printing · tracking entry · disputes · questions inbox · read-only catalog/stock/price views with listing status and rejection reasons · performance + score detail · transactions/settlement · branches + users · integration status.
- **Client back-office:** seller management + provisioning · catalog moderation queue (match candidates, new products, enrichment) · listing moderation · category tree + per-category attributes · brand registry · synonym dictionary · service areas + postal ranges + regions · allocation + price-guard audit · score configuration · commission configuration · campaigns + homepage · dispute arbitration · integration observability (events, failures, replay) · reports.

**Known engineering flags (raise these, don't discover them in sprint two):** the kit definition/pricing boundary vs the Hub (FR-CAT-045 ↔ FR-LST-001/FR-PRC-001, already flagged to the client) · FR-GEO-004 statefulness at scale — prototype in month one · match-threshold calibration against real seller feeds (moderation volume depends on it) · gateway split confirmation per payment method before the payment workstream starts.